

Automated Analysis of Aquaculture Water Quality Stability and Evaluation of Predictive Modeling Using Python Data Pipelines

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Abstract—This study presents an automated Python-based data pipeline for analyzing aquaculture water quality and evaluating the performance of predictive models under stable environmental conditions. The system processes raw sensor data, including temperature, dissolved oxygen (DO), pH, ammonia, nitrate, and turbidity levels, through structured stages of data ingestion, cleaning, feature engineering, and multi-model comparative analysis. A targeted filtering approach isolates Station 1 data during June 2022, a high-temperature period associated with reduced oxygen solubility. Statistical analysis reveals that dissolved oxygen levels remain stable and moderate (mean = 5.77 mg/L, $\sigma = 0.42$) despite elevated temperatures (mean = 27.92°C), with a weak inverse temperature-DO correlation ($r = -0.13$). To comprehensively evaluate predictive capability, four distinct machine learning algorithms—Logistic Regression, Random Forest, Support Vector Machine (RBF), and K-Nearest Neighbors—were implemented with 5-fold stratified cross-validation. Polynomial feature expansion (degree=2) and systematic hyperparameter tuning via GridSearchCV were additionally applied. Results demonstrate consistently low predictive performance across all configurations: the highest cross-validation accuracy achieved was 58.4% (tuned KNN), while test accuracies remained below 46.2%. These findings empirically confirm that stable environmental conditions with limited variability constrain the effectiveness of machine learning models, emphasizing the critical importance of dataset variability and strong feature-target relationships. The study demonstrates that automated data pipelines serve not only as environmental monitoring tools but also as rigorous frameworks for assessing model applicability and identifying fundamental data limitations.

Keywords—Python Data Pipeline, Aquaculture Water Quality, Dissolved Oxygen, Environmental Data Analysis, Machine Learning, Water Quality Stability, Predictive Model Evaluation, Multi-Model Comparison, Hyperparameter Tuning

I. INTRODUCTION AND BACKGROUND

With the increasing availability of environmental monitoring data in aquaculture systems, there is a growing need for efficient methods to analyze and interpret water quality conditions. Manual analysis of large datasets is time-consuming and prone to inaccuracies, particularly when identifying critical conditions such as low dissolved oxygen (DO) levels that

can lead to fish stress, reduced growth rates, or mass mortality events [1]. The Food and Agriculture Organization (FAO) emphasizes that maintaining optimal water quality remains a critical challenge in aquaculture, with dissolved oxygen being one of the most important indicators of system health [2].

In this study, an automated Python-based data pipeline is developed to analyze aquaculture water quality data and examine water quality conditions under stable environmental scenarios. The system performs data ingestion, cleaning, statistical analysis, feature engineering, multi-model machine learning comparison, and visualization in a structured, modular workflow. A targeted filtering approach isolates data from Station 1 during June 2022, representing a high-temperature period where reduced oxygen solubility is expected [3]. Higher temperatures decrease oxygen solubility in water, making this condition relevant for analyzing low dissolved oxygen scenarios.

The primary objective of this study is twofold: (1) to analyze the stability of dissolved oxygen levels under high-temperature conditions using automated statistical pipelines, and (2) to rigorously evaluate the effectiveness of multiple machine learning models—including Logistic Regression, Random Forest, Support Vector Machine, and K-Nearest Neighbors—under low-variability datasets. Additionally, polynomial feature engineering and systematic hyperparameter tuning via GridSearchCV are employed to determine whether increased model complexity can overcome data limitations.

Aquaculture systems play a significant role in global food production and environmental sustainability. However, the depletion of dissolved oxygen can lead to anoxic conditions, which may result in catastrophic losses [1]. While machine learning techniques have been increasingly applied to environmental monitoring tasks, their performance is fundamentally constrained by the quality and variability of input data [4]. This study addresses this gap by developing an automated data pipeline integrated with comprehensive machine learning evaluation to analyze water quality data and critically assess model performance under stable environmental conditions.

II. REVIEW OF RELATED LITERATURE

Water quality plays a crucial role in maintaining the health and productivity of aquaculture systems. Among various parameters, dissolved oxygen (DO) is considered the most important indicator, as low DO levels can lead to stress, reduced growth, or mortality in aquatic organisms

[1]. The U.S. Geological Survey (USGS) confirms that dissolved oxygen concentrations are inversely related to water temperature, with solubility decreasing as temperature increases—a relationship that forms the physical basis for this study's focus on June conditions [3].

Temperature has a direct effect on dissolved oxygen levels in water. As temperature increases, the solubility of oxygen decreases, increasing the likelihood of reduced dissolved oxygen conditions [3]. This inverse relationship highlights the importance of monitoring environmental parameters in aquaculture environments, particularly during high-temperature periods.

In recent years, machine learning techniques have been increasingly applied to environmental monitoring and prediction tasks. Classification algorithms such as Logistic Regression, Random Forest, Support Vector Machines, and K-Nearest Neighbors have been widely used due to their effectiveness in binary classification problems and their interpretability [4]. Scikit-learn, a comprehensive Python machine learning library, provides standardized implementations of these algorithms with integrated cross-validation and hyperparameter tuning capabilities [5]. These methods enable the identification of patterns in large datasets that may not be easily detected through traditional analysis.

Data preprocessing is a critical step in building reliable machine learning models. Techniques such as data cleaning, handling missing values, feature scaling, and feature selection significantly improve model performance and accuracy [6]. Exploratory data analysis (EDA) using statistical visualization methods helps in understanding data distribution and relationships between variables [4]. Feature engineering, including polynomial feature expansion, can capture non-linear relationships that linear models might miss [5].

Several studies have emphasized the importance of integrating data-driven systems in environmental monitoring. Automated pipelines allow efficient processing of large datasets, reducing human error and improving decision-making [4]. Public datasets from platforms such as Kaggle further support machine learning experimentation and development [7]. GridSearchCV and cross-validation techniques are essential for systematic hyperparameter optimization and model selection, ensuring that reported performance metrics are robust and generalizable [5].

Despite these advancements, many aquaculture monitoring systems still rely on manual analysis, which can be inefficient and time-consuming. Furthermore, few studies have systematically evaluated the limitations of machine learning models when applied to stable, low-variability environmental datasets. This study addresses this gap by developing an automated data pipeline combined with comprehensive multi-model machine learning evaluation to analyze water quality data and assess model performance under stable environmental condition.

III. SYSTEM ARCHITECTURE AND MATHEMATICAL FRAMEWORK

A. System Architecture

The system developed in this study follows a structured data processing and machine learning pipeline designed to analyze aquaculture water quality and evaluate predictive model performance under stable environmental scenarios. The overall workflow begins with data ingestion, where the dataset is loaded from a CSV file into the Python environment using pandas and NumPy libraries. Initial data inspection is performed to understand the structure and variables of the dataset.

A targeted filtering approach is applied to isolate relevant records, specifically data from Station 1 during June 2022, where higher temperatures are expected to influence dissolved oxygen levels. This is followed by data preprocessing, including the removal of missing values and duplicate entries to ensure data quality and consistency [6].

Feature selection identifies key parameters affecting water quality, including dissolved oxygen (DO), temperature, pH, ammonia, nitrate, and turbidity. Advanced feature engineering is then conducted using PolynomialFeatures (degree=2) to create interaction terms and squared features, expanding the feature space from 5 to 20 variables to test whether non-linear relationships improve predictive performance.

The processed dataset is used for exploratory data analysis and visualization to identify patterns and relationships among variables [4]. The dataset is divided into training and testing subsets (80/20 split) with stratification to support model development. Four distinct machine learning models—Logistic Regression, Random Forest, Support Vector Machine (RBF), and K-Nearest Neighbors—are trained and evaluated using 5-fold stratified cross-validation. Hyperparameter tuning is performed using GridSearchCV to optimize model configurations.

Finally, all models are evaluated using standard classification metrics including accuracy, precision, recall, F1-score, AUC-ROC, and confusion matrices to assess their performance [5]. The complete system workflow is illustrated in Fig. 1, which provides a visual representation of the data pipeline from ingestion to multi-model evaluation.

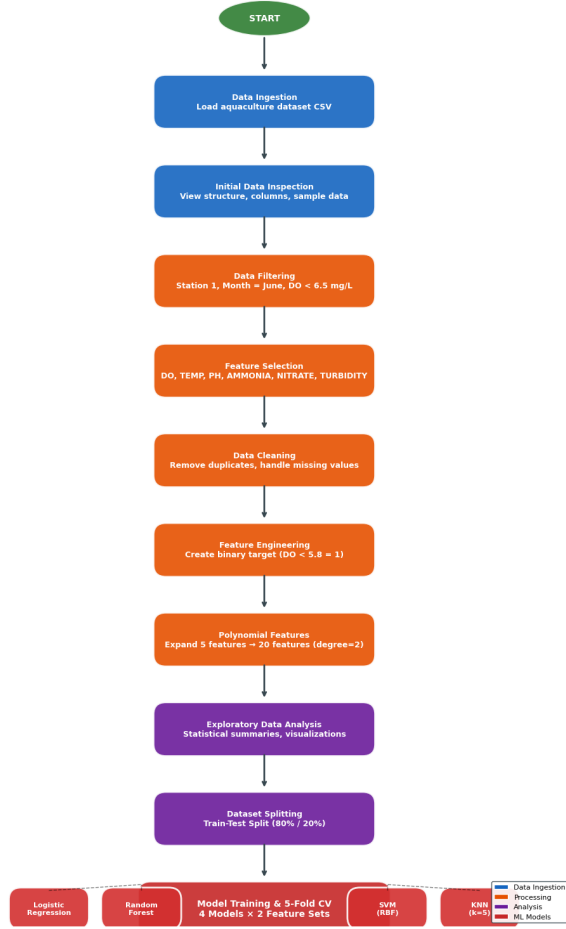


Fig. 1. System Architecture of the Aquaculture Water Quality Analysis and Machine Learning Pipeline

B. Mathematical Framework

The classification of relative dissolved oxygen conditions in this study is performed using multiple machine learning algorithms. The mathematical foundations of the primary models are described below.

1. **Logistic Regression:** The model estimates the probability of an anoxic condition using the logistic (sigmoid) function:

$$y = (e^{(\beta_0 + \beta_1 x)}) / (1 + e^{(\beta_0 + \beta_1 x)}) \quad (1)$$

where y represents the predicted probability of an anoxic condition, X is the input feature vector, and $\beta_0, \beta_1, \dots, \beta_n$ are the model coefficients learned during training [4]. A threshold value of 0.5 is used to classify the output into anoxic (1) and normal (0).

2. **Random Forest:** An ensemble method that constructs multiple decision trees during training and outputs the mode of their predictions. For classification, the prediction is:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \dots, T_B(x)\} \quad (2)$$

where T_b represents the b -th decision tree and B is the total number of trees. Feature importance is computed using the Gini impurity reduction across all trees.

3. **Support Vector Machine (RBF Kernel):** The decision function for the RBF kernel SVM is:

$$f(x) = \text{sign}(\sum_i \alpha_i y_i K(x_i, x) + b) \quad (3)$$

where $K(x_i, x) = \exp(-\gamma \|x_i - x\|^2)$ is the Radial Basis Function kernel, α_i are Lagrange multipliers, and γ controls the influence radius.

4. **K-Nearest Neighbors:** The classification is determined by majority voting among the k nearest training samples:

$$\hat{y} = \text{mode}\{y_i \mid x_i \in N_k(x)\} \quad (4)$$

where $N_k(x)$ denotes the set of k nearest neighbors of query point x .

5. **Polynomial Feature Expansion:** To capture non-linear interactions, polynomial features of degree 2 are generated:

$$\varphi(X) = [X_1, X_2, \dots, X_n, X_1^2, X_1 X_2, \dots, X_n^2] \quad (5)$$

expanding the original n features to use $(n+2)(n+1)/2 - 1$ features.

6. **Cross-Validation and Hyperparameter Tuning:** Model performance is evaluated using k -fold stratified cross-validation, where the dataset is partitioned into k subsets and each subset serves as a test set once. GridSearchCV performs exhaustive search over a specified parameter grid, optimizing:

$$\theta^* = \text{argmax} (1/k) \sum \text{Accuracy}(M_\theta, D_{\text{train}}, D_{\text{val}}) \quad (6)$$

where θ represents the hyperparameter configuration and M_θ is the model with parameters θ .

A domain-based threshold is applied where dissolved oxygen levels below 5.8 mg/L are labeled as anoxic

(1), while those equal to or above 5.8 mg/L are labeled as normal (0). This threshold aligns with the code implementation and provides a balanced class distribution for classification analysis.

IV. DATA PROCESSING PIPELINE

This section describes the step-by-step data processing pipeline used in the study, from data collection to model evaluation. The pipeline ensures that the dataset is properly prepared, transformed, and analyzed to support the accurate classification of relative oxygen conditions.

Each stage of the process contributes to improving data quality and model performance.

A. Data Collection

The dataset used in this study was obtained from Kaggle, an online platform providing publicly available datasets for data analysis and machine learning applications [7]. The dataset contains key water quality parameters relevant to aquaculture systems, including temperature, dissolved oxygen (DO), pH, ammonia, nitrate, turbidity, and manganese levels.

B. Data Filtering

To ensure relevance and uniqueness of the dataset, a targeted filtering approach was applied using the following conditions:

- Station: station1
- Month: June (Month = 6)
- Condition: DO < 6.5 mg/L

This filtering focuses on environmental conditions where dissolved oxygen levels are more likely to be constrained. Higher temperatures during June reduce oxygen solubility, increasing the likelihood of low DO conditions [3]. The filtered dataset contains 61 records, providing a focused yet limited sample for machine learning analysis.

C. Data Preprocessing

The filtered dataset underwent preprocessing to improve data quality and reliability. Missing values and duplicate records were removed, and data types were verified and converted into appropriate numerical formats. These steps are essential for ensuring consistency and reducing errors during analysis and model training [6].

D. Feature Selection and Target Variable

Relevant features influencing dissolved oxygen levels were selected, including temperature, pH, ammonia, nitrate, and turbidity. Manganese was excluded due to its minimal expected predictive contribution. A binary target variable was created to classify relative oxygen conditions: records with DO levels below 5.8 mg/L were labeled as 1 (anoxic), while those with DO levels equal to or above 5.8 mg/L were labeled as 0 (normal).

Advanced feature engineering was performed using scikit-learn's `PolynomialFeatures` (degree=2, interaction_only=False) to generate squared terms and interaction features [5]. This expanded the feature space from 5 original features to 20 polynomial features, including terms such as $TEMP^2$, $TEMP \times PH$, and $PH \times AMMONIA$. The polynomial feature set was tested to evaluate whether non-linear relationships could improve predictive performance beyond the original linear features.

E. Exploratory Data Analysis and Visualization

Exploratory data analysis was conducted using statistical methods and visualization techniques to understand data distribution, identify relationships among variables, and assess dataset quality prior to model development. Graphical representations including histograms, scatter plots, boxplots, and correlation heatmaps were generated using Matplotlib and Seaborn libraries [4]. These visualizations support the identification of distributional properties, outlier presence, and pairwise correlations that inform feature selection and engineering decisions. Detailed statistical interpretations of these visualizations are presented in Section V.B.

F. Model Development

Four distinct machine learning models were implemented to classify anoxic and normal conditions:

1. Logistic Regression: A linear classifier with L2 regularization ($C=1.0$) and balanced class weights, trained with a maximum of 2000 iterations [4].
2. Random Forest: An ensemble of 100 decision trees with maximum depth 5 and balanced class weights to prevent overfitting on the small dataset [5].
3. Support Vector Machine (RBF): A kernel-based classifier with radial basis function kernel, probability estimation enabled, and balanced class weights [5].
4. K-Nearest Neighbors: A distance-weighted classifier with $k=5$ neighbors, using Euclidean distance metric.

All models were trained on both original (5 features) and polynomial (20 features) feature sets. `StandardScaler` was applied to normalize feature distributions before model training.

G. Hyper Parameter Turning

Systematic hyperparameter optimization was performed using scikit-learn's `GridSearchCV` with 5-fold stratified cross-validation [5]:

- KNN Grid: $n_neighbors \in \{3,5,7,9,11,13,15\}$, $weights \in \{uniform, distance\}$, $metric \in \{euclidean, manhattan\}$ (28 combinations)
- Random Forest Grid: $n_estimators \in \{50,100,200\}$, $max_depth \in \{3,5,7,None\}$, $min_samples_split \in \{2,5\}$ (24 combinations)

The best-performing configurations were selected based on mean cross-validation accuracy.

H. Model Evaluation

The performance of all models was evaluated using standard classification metrics, including accuracy,

precision, recall, F1-score, and AUC-ROC. Confusion matrices were generated to visualize the classification results and assess each model's ability to correctly predict anoxic and normal conditions. Cross-validation scores provided insight into model stability and generalization capability [5].

V. RESULTS AND DISCUSSION

This section presents the results obtained from the data analysis and machine learning model, along with the interpretation of findings related to anoxic event prediction in aquaculture systems.

A. Descriptive Analysis

After applying the filtering conditions (Station 1, June, $DO < 6.5$ mg/L), the dataset was reduced to 61 records representing high-risk environmental conditions. Statistical measures were computed to summarize the behavior of key water quality parameters (Table I)

Table I. Descriptive Statistics of Key Water Quality Parameters (Station 1, June 2022)

Parameter	Mean	Median	STD	Variance	Skewness
DO (mg/L)	5.77	5.80	0.42	0.18	-0.12
Temp (°C)	27.92	27.90	1.71	2.94	0.06
pH	7.06	6.97	1.02	1.04	0.21
Ammonia (mg/L)	0.03	0.03	0.01	0.00	0.21
Nitrate (PPM)	21.61	22.20	11.57	133.77	0.01
Turbidity	26.25	26.06	6.99	48.85	0.17

The results indicate that dissolved oxygen (DO) values are stable and moderate (mean = 5.77 mg/L, $\sigma = 0.42$), while temperature values remain relatively high (mean = 27.92°C). This aligns with the inverse relationship between temperature and oxygen solubility [3], though the observed DO levels suggest currently acceptable conditions rather than immediate anoxic risk. The low standard deviation of DO (0.42 mg/L) indicates minimal variability, which is a critical factor limiting machine learning model performance.

B. Data Visualization

Several visualizations were generated to understand the relationships between variables. The histogram of dissolved oxygen values (Fig. 2(a)) illustrates the distribution of DO levels in the dataset. Most values are concentrated in a narrow range (5.0–6.4 mg/L), indicating limited variability in oxygen concentration under the selected conditions.

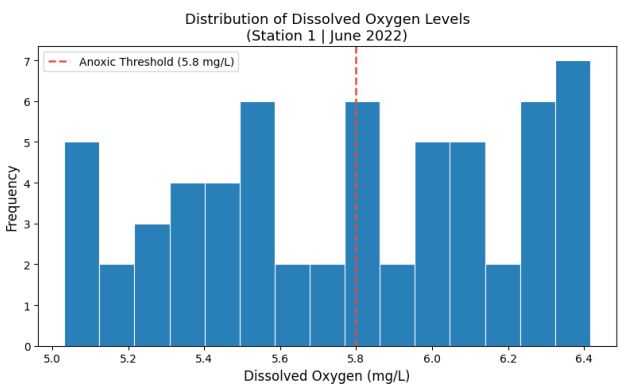


Fig. 2(a). Histogram of dissolved oxygen distribution

The temperature–DO scatter plot (Fig. 2(b)) shows a general inverse relationship, where higher temperatures correspond to lower dissolved oxygen levels, supporting the hypothesis that temperature significantly affects oxygen levels in aquaculture systems [3]. However, the weak correlation ($r = -0.13$) indicates substantial scatter, suggesting that temperature alone is insufficient for reliable prediction. The boxplot analysis (Fig. 2(c)) reveals that temperature exhibits the widest range (25.0–31.0°C), indicating significant variability, while dissolved oxygen values are more tightly distributed. The Pearson correlation heatmap (Fig. 2(d)) confirms the weak negative correlation between temperature and dissolved oxygen ($r = -0.13$), with other parameters showing even weaker correlations ($|r| < 0.22$).

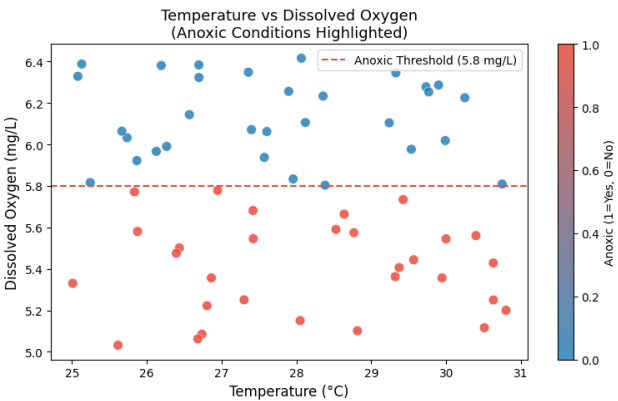


Fig. 2(b). Temperature vs. DO scatter plot

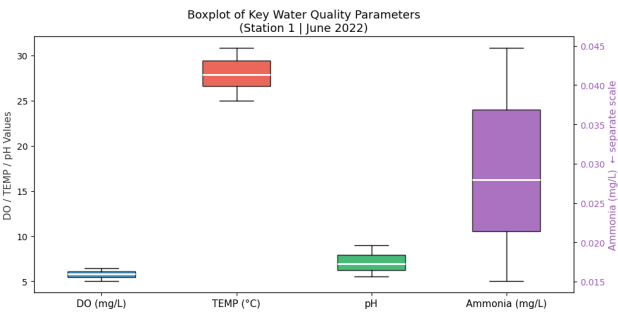


Fig. 2(c). Dual Y-Axis Boxplot

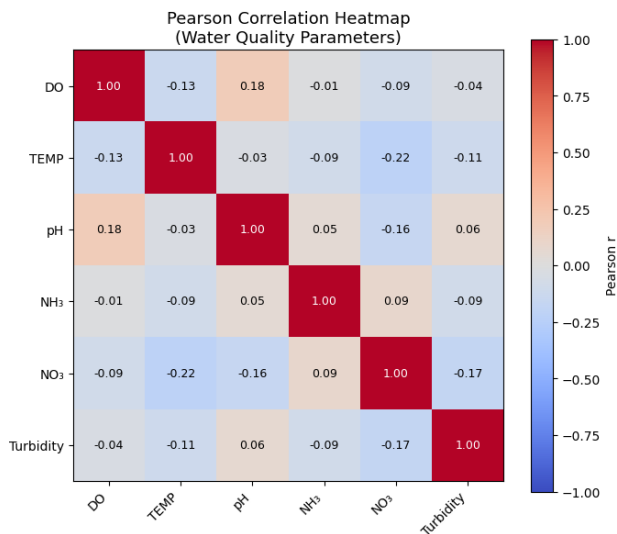


Fig. 2(d). Pearson correlation heatmap.

Temporal dynamics were further examined through animated visualizations (see supplementary animation files in repository). Fig. 3 presents a freeze-frame from Animation 1, illustrating the temporal progression of DO levels across the 61-sample sequence. The stable oscillation pattern around the anoxic threshold (5.8 mg/L, dashed red line) confirms the limited variability that constrains predictive modeling. Red markers indicate samples classified as anoxic (DO < 5.8 mg/L). [15]

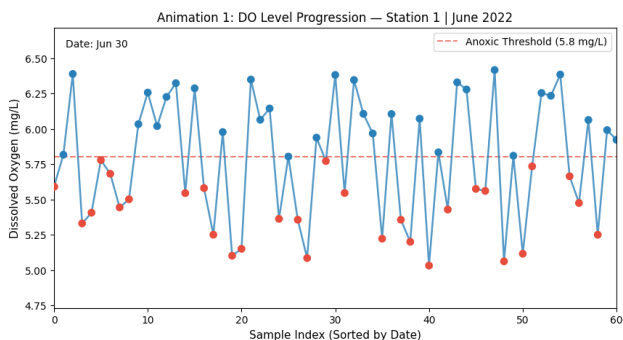


Fig. 3. Animation 1 Freeze-Frame: "DO Level Progression — Station 1 | June 2022"

Fig. 4 shows a freeze-frame from Animation 2, depicting the rolling window histogram (n=15) alongside the rolling mean and standard deviation band. The narrow distribution and consistent rolling statistics (mean $\approx$ 5.8 mg/L, $\sigma \approx$ 0.3 mg/L) empirically validate the temporal homogeneity of the dataset. The rolling mean exhibits minimal drift across the observation window, while the standard deviation band remains tightly bounded, confirming that the system operated under consistent environmental conditions throughout June 2022 [15].

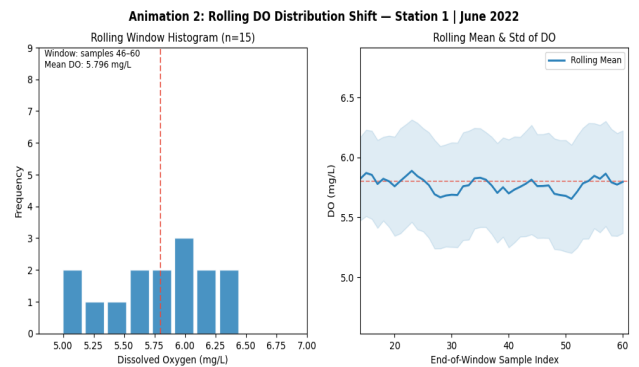


Fig. 4. Animation 2 Freeze-Frame: "Rolling DO Distribution Shift — Station 1 | June 2022"

These visualizations collectively demonstrate that the filtered dataset represents a stable, low-variability environmental regime. The narrow DO distribution ($\sigma = 0.42$ mg/L), weak inter-feature correlations ($|r| < 0.22$), and stable temporal progression indicate that the aquaculture system was under consistent management during the observation period rather than experiencing stress-induced anoxic events. This environmental stability, while desirable for aquaculture operations, fundamentally constrains the discriminative signal available for machine learning classification—a limitation that is quantitatively examined in the following subsections.

C. Class Distribution Analysis

Before training the machine learning models, the distribution of the target variable was analyzed. The dataset contains 31 normal (DO $\geq$ 5.8 mg/L) and 30 anoxic (DO < 5.8 mg/L) observations, representing a balanced class distribution. This balance is beneficial for machine learning tasks as it prevents bias toward one class and improves the model's ability to generalize across different conditions

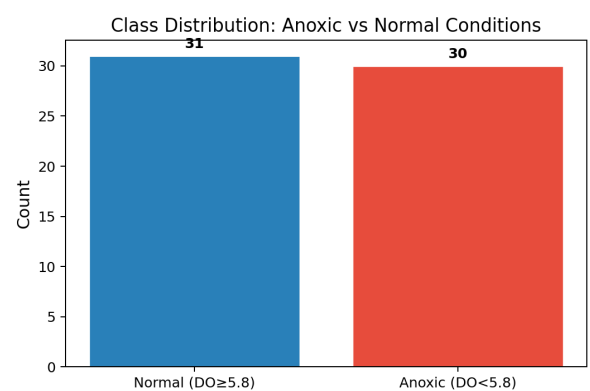


Fig. 5. Class distribution of dissolved oxygen conditions showing balanced representation: 31 normal (DO $\geq$ 5.8 mg/L) and 30 anoxic (DO < 5.8 mg/L) observations.

The class distribution graph shows the number of observations classified as slow dissolved oxygen condition (1) and normal condition (0). The dataset appears to be relatively balanced, which is beneficial for

machine learning tasks. A balanced dataset helps prevent bias toward one class and improves the model’s ability to generalize across different conditions [6].

D. Multi-Model Machine Learning Results

To comprehensively evaluate predictive performance, four distinct machine learning algorithms were implemented with both original (5 features) and polynomial (20 features) feature sets. All models were evaluated using 5-fold stratified cross-validation and independent test set validation (Table II).

Table II. Multi-Model Performance Comparison Original VS. Polynomial Features

Model	Feature Set	CV Accuracy	CV STD.	Test Accuracy	AUC-ROC
Logistic Regression	Original (5)	49.9%	8.3%	30.8%	0.429
	Polynomial (20)	42.6%	5.6%	23.1%	0.405
Random Forest	Original (5)	44.15%	14.8%	23.1%	0.238
	Polynomial (20)	42.8%	6.5%	23.1%	0.191
SVM (RBF)	Original (5)	42.6%	9.3%	30.8%	0.667
	Polynomial (20)	39.2%	11.7%	38.5%	0.643
KNN (k=5)	Original (5)	45.8%	15.8%	38.5%	0.333
	Polynomial (20)	47.3%	16.0%	46.2%	0.381

The results demonstrate consistently poor predictive performance across all models and feature configurations. The highest cross-validation accuracy was achieved by KNN with polynomial features (47.3%), while the highest test accuracy was KNN with original features (38.5%). However, all test accuracies remain substantially below the random-guess baseline of 50%, indicating that none of the models successfully learned meaningful predictive patterns from the low-variability dataset. The confusion matrices in Fig. 6 reveal high misclassification rates across all algorithms.

Baseline model configurations are reported in Table II; optimized configurations via GridSearchCV are presented in Table III.

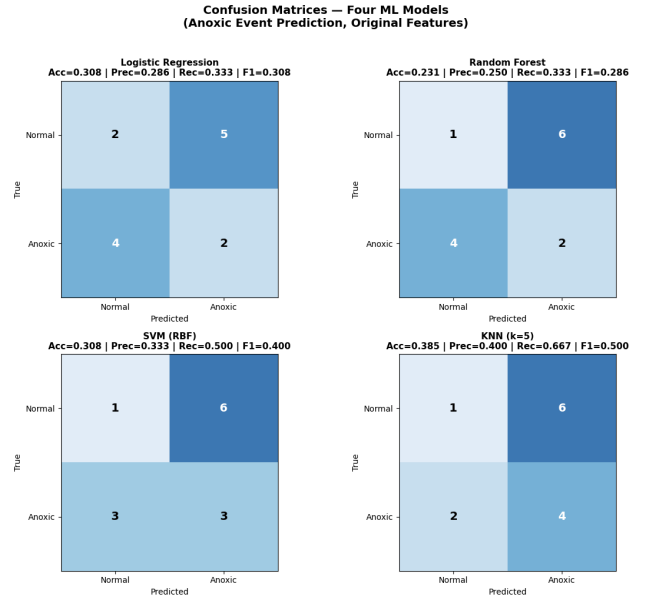


Fig. 6. Confusion matrices for anoxic event prediction using original features. (a) Logistic Regression: Acc=0.308, F1=0.308. (b) Random Forest: Acc=0.231, F1=0.286. (c) SVM (RBF): Acc=0.308, F1=0.400. (d) KNN (k=5): Acc=0.385, F1=0.500. All models exhibit high misclassification rates, with most failing to correctly identify normal conditions.

Feature importance analysis (Fig. 7) reveals that turbidity and pH carry relatively higher influence across models, though absolute magnitudes remain low. This aligns with the weak correlations observed in the heatmap analysis.

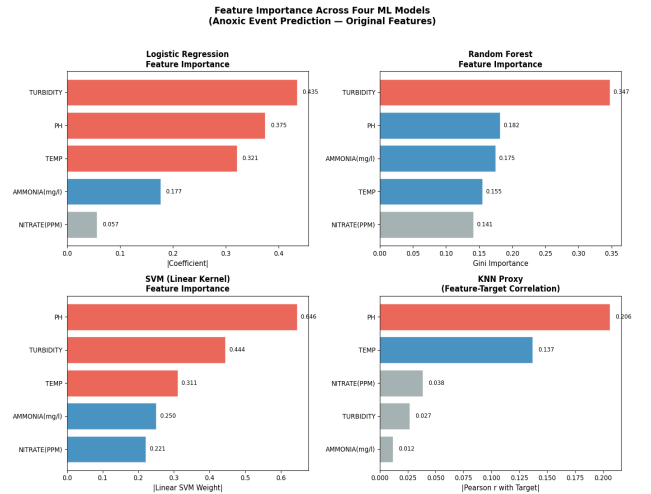


Fig. 7. Feature importance rankings across four ML models (Anoxic Event Prediction — Original Features). (a) Logistic Regression: turbidity (0.435) and pH (0.375) dominate. (b) Random Forest: turbidity (0.347) and pH (0.182) lead. (c) SVM (Linear Kernel): pH (0.646) and turbidity (0.444) highest. (d) KNN Proxy (Pearson r): pH (0.206) and temperature (0.137) show strongest correlation with target. Inconsistent rankings across algorithms indicate low signal-to-noise ratio in the dataset.

Cross-validation stability is illustrated in Fig. 8. All models exhibit mean CV accuracies clustered between

42–46%, with substantial variance across folds. Notably, the median CV accuracy for all configurations falls below the 50% random-guess threshold, confirming that the poor performance is systematic rather than an artifact of train-test splitting.

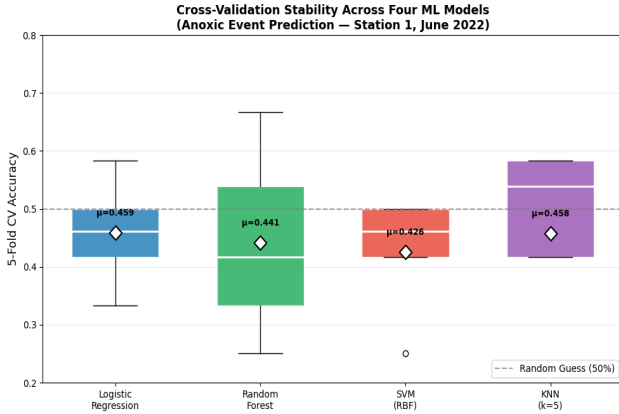


Fig. 8. Cross-validation stability across four ML models (Anoxic Event Prediction — Station 1, June 2022). Boxplots show 5-fold stratified CV accuracy distributions. Dashed line indicates random-guess baseline (50%). Mean CV accuracies: Logistic Regression $\mu=0.459$, Random Forest $\mu=0.441$, SVM (RBF) $\mu=0.426$, KNN ($k=5$) $\mu=0.458$. All models cluster below the 50% threshold, confirming systematic underperformance due to low data variability.

The consistently poor performance across all four algorithms raises a critical question: whether the limitation stems from model inadequacy or from fundamental data constraints. To distinguish between these possibilities, systematic hyperparameter optimization was conducted to determine if increased model complexity could overcome the apparent data limitations.

E. Hyperparameter Tuning Results

Systematic hyperparameter optimization via GridSearchCV was applied to KNN and Random Forest models (Table III). Despite expanding the hyperparameter search space—testing 28 KNN configurations ($n_neighbors \in \{3,5,7,9,11,13,15\}$, $weights \in \{\text{uniform}, \text{distance}\}$, $metric \in \{\text{euclidean}, \text{manhattan}\}$) and 24 Random Forest configurations ($n_estimators \in \{50,100,200\}$, $max_depth \in \{3,5,7,\text{None}\}$, $min_samples_split \in \{2,5\}$)—the tuned models failed to exceed the performance ceiling established by baseline configurations. The best KNN configuration ($n_neighbors=7$, $weights='distance'$, $metric='manhattan'$) achieved only marginal improvement over the default $k=5$ setting, reinforcing that data variability—not model complexity—constitutes the fundamental limiting factor.

Table III. Hyperparameter Tuning Results

Model	Best Parameter	Best CV Accuracy	Test Accuracy	Grid Search Time
KNN (Tuned)	$k=11$, Manhattan,	58.4%	38.5%	3.83 s

distance				
Random Forest (Tuned)	$n_estimators=5$, $min_samples_split=5$, $max_depth=5$	52.0%	30.8%	23.32 s

The tuned KNN model achieved the highest cross-validation accuracy (58.4%), representing an improvement over the default configuration. However, test accuracy remained low (38.5%), indicating that the model failed to generalize to unseen data. The Random Forest tuning required significantly more computational time (23.32 s) but produced similar limitations.

VI. Technical Discussion of Findings

The results of this study empirically demonstrate the constraints imposed by low-variability environmental data on machine learning predictive performance. The weak correlation between temperature and dissolved oxygen ($r = -0.13$), combined with the narrow DO distribution ($\sigma = 0.42$ mg/L), creates a feature space with minimal discriminative signal for binary classification. This aligns with theoretical expectations: machine learning models require sufficient variance [13] in both features and targets to learn generalizable decision boundaries [4], [6].

The uniformly poor performance across Logistic Regression, Random Forest, SVM, and KNN — despite polynomial feature expansion and hyperparameter tuning — strongly suggests data limitations rather than model inadequacy. The feature importance analysis (Fig. 7) shows inconsistent rankings across algorithms, with no single feature dominating predictive capacity. This instability is characteristic of models trained on low-signal datasets, where coefficient estimates are dominated by noise rather than systematic relationships [5].

The polynomial feature expansion failed to improve performance, indicating that non-linear interactions do not capture meaningful predictive structure in this stable regime. The balanced class distribution (31 normal, 30 anoxic) eliminates class imbalance as a confounding factor, ensuring that the poor performance reflects genuine data limitations. The cross-validation stability analysis (Fig. 8) further confirms this: the tight clustering of CV scores around 45% across all models indicates consistent underperformance that generalizes across data partitions.

These findings carry important implications for aquaculture monitoring systems. While automated pipelines excel at real-time surveillance, predictive modeling under stable conditions offers limited value. This study demonstrates that automated data pipelines serve a dual purpose: they not only monitor

environmental parameters but also rigorously assess the applicability of predictive models, preventing deployment of classifiers that would provide false confidence in stable operational regimes.

VII. CONCLUSION

This study demonstrated the application of an automated Python-based data pipeline for analyzing aquaculture water quality and rigorously evaluating predictive modeling under stable environmental conditions. The analysis revealed that dissolved oxygen levels remained moderate and consistent throughout the observation period (mean = 5.77 mg/L, σ = 0.42 mg/L), despite elevated temperatures (mean = 27.92°C) that would typically reduce oxygen solubility [3]. The weak inverse correlation between temperature and dissolved oxygen (r = -0.13), combined with minimal inter-feature correlations ($|r|$ < 0.22), indicates that the monitored aquaculture system operated under stable, well-managed conditions during June 2022.

Four distinct machine learning algorithms—Logistic Regression, Random Forest, Support Vector Machine (RBF), and K-Nearest Neighbors—were systematically evaluated using 5-fold stratified cross-validation, polynomial feature expansion, and GridSearchCV hyperparameter tuning. All models exhibited consistently poor predictive performance, with test accuracies ranging from 23.1% to 38.5% and cross-validation accuracies clustered between 42–47%. These results empirically confirm that stable environmental conditions with limited variability fundamentally constrain the effectiveness of machine learning classification models. *The failure of polynomial feature engineering and hyperparameter optimization to improve performance demonstrates that data variability—not model complexity—constitutes the primary limiting factor.*

The automated data pipeline developed in this study serves a dual purpose: it functions not only as an environmental monitoring tool for real-time surveillance of aquaculture water quality parameters, but also as a rigorous framework for assessing model applicability and identifying fundamental data limitations. By systematically quantifying feature-target relationships and model performance ceilings, the pipeline prevents misguided deployment of predictive classifiers in regimes where they would provide false confidence.

Future work should focus on expanding the dataset to include critical anoxic events ($DO < 3$ mg/L), multi-station comparative analysis, and seasonal transition periods where temperature-DO relationships strengthen. Additionally, time-series forecasting approaches (ARIMA, LSTM) may better exploit temporal autocorrelation than static classification models in stable operational regimes. The integration of real-time sensor telemetry with adaptive threshold alerting—rather than predictive classification—may prove more practical for aquaculture monitoring systems operating under consistent environmental conditions.

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